Manav
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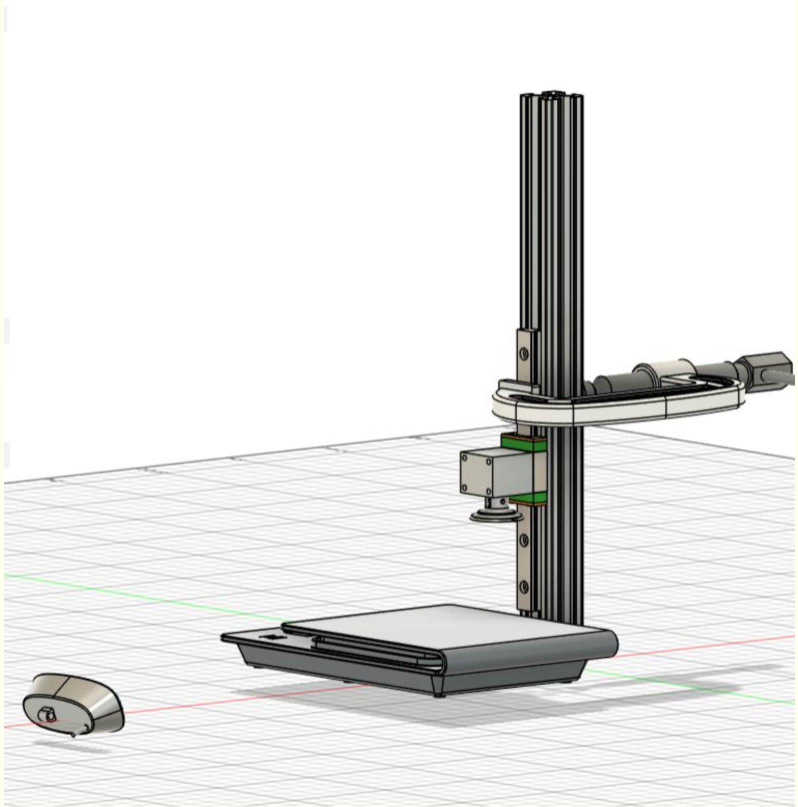
Gebelt Climbing Robot- Week 2

Testbed

- Manual push and pull while recording the scale data
  - Force to bottom out the cup
  - Force to remove the cup
  - Shear???

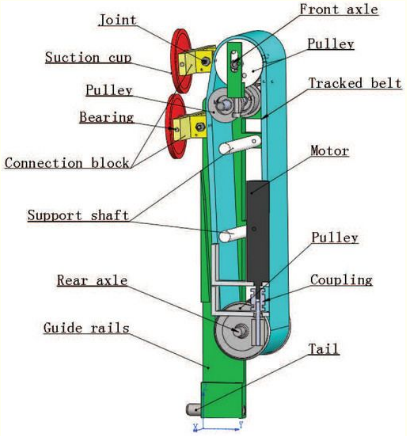
- Getting deformation data from the camera will be helpful
  - Camera video of cup deformation so we can correlate push-in force → seal quality → pull-off strength → vibration/stability
  - Start scale reading and camera at same time

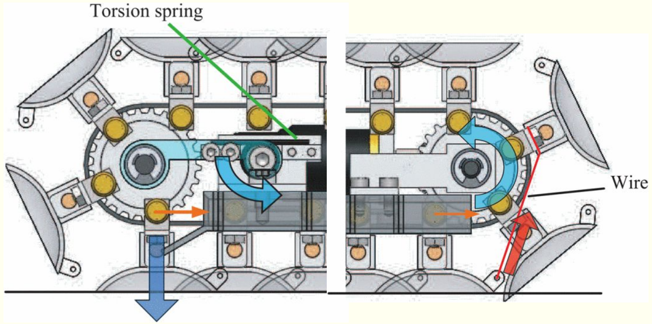


Item	Vendor	Part Number	Cost	Qty	Total	URL	Notes	Status
Suction Cups								
Vacuum Cup Height -Adjustable, Push-on, Single	McMaster-Carr	7275A23	$11.34	1	$11.34	https://www.mcm	Active suction cu	Ready to Orc
Vacuum Cup Height -Adjusting, Push-on, Single I	McMaster-Carr	5427A106	$4.95	1	$4.95	https://www.mcm	Buna-N rubber n	Ready to Orc
T-Slotted Framing Single Four Slot Rail, Silver, 1" High	McMaster-Carr	47065T411	$8.67	1	$8.67	https://www.mcm	Aluminum extrus	Ready to Orc
Suction-Cup Hooks 1-1/8" Wide x 1-1/8" High x 1/2" De	McMaster-Carr	65825A3	$11.64	1	$11.64	https://www.mcm	Has hooks(we ca	Ready to Orc

Next Steps

- Conduct testing next week
    - After seeing forces and deformation data we can consider the number of the suction cups we need, per-cup strength, the roughness of the target surface, and the cup's softness/stiffness
    - Make a decision for initial prototype (active or passive suction)
    - Begin researching, ideating, and designing the mechanism





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Gebelt Climbing Robot- Week 3

Testbed

- Changed testbed to use new linear rail
  - New method of pushing and pulling so that normal force can be controlled
- Shear Force:
  - Thorlabs xy plane will be our bed and the same setup can be used, we will just move the bed until we get a slip
  - Press Fn up to max value, gotten from FN only test
  - Apply tangential motion my moving xy plane
  - Stop when a slip happens

- Only Normal force:
  - Start DAQ
  - Apply normal force up to X N
  - Unscrew
  - Repeat until the force does not change a lot
    - This tells us how much force we need to get the best seal, also our pulling force will be recorded every time

Sensing

- Figured out how to connect to IP
- Log force–torque data from the Nordbo sensor and video from a USB webcam
    - Press S to start recording
        - Script captures video frames and stores averaged force–torque samples in an array (.npz)
    - Press Q to stop recording and save both the video and the force–torque data

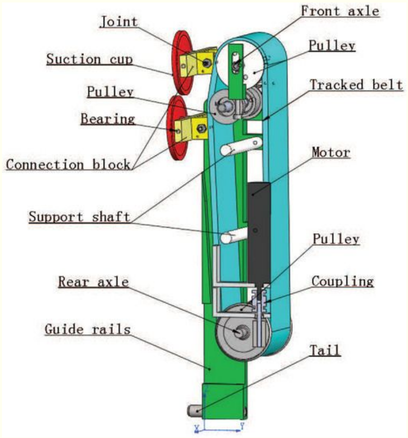
```python
def main():
    while True:
        ret, frame = cap.read()
        if not ret:
            break

        recvData = recvMsg()
        Fx = struct.unpack('!d', recvData[2:10])[0]
        Fy = struct.unpack('!d', recvData[10:18])[0]
        Fz = struct.unpack('!d', recvData[18:26])[0]
        Tx = struct.unpack('!d', recvData[26:34])[0]
        Ty = struct.unpack('!d', recvData[34:42])[0]
        Tz = struct.unpack('!d', recvData[42:50])[0]

        if recording:
            out.write(frame)

            data_buffer.append([Fx, Fy, Fz, Tx, Ty, Tz])

            # Average every 10 samples
            if len(data_buffer) == 10:
                avg = np.mean(data_buffer, axis=0)
                averaged_data.append(avg)
                data_buffer = []

        cv2.imshow("Webcam", frame)

        key = cv2.waitKey(1) & 0xFF
        if key == ord('s') and not recording:
            print("Started recording")
            recording = True
        elif key == ord('q') and recording:
            print("Stopped recording")
            break

    # Save FT data
    np.savez(npz_filename, data=np.array(averaged_data))
    print(f"Force-torque data saved at {npz_filename}")
    print(f"Video saved at {video_filename}")

    sendData = '03' + CMD_TYPE_SENSOR_TRANSMIT + SENSOR_TRANSMIT_TYPE_STOP
    s.send(bytearray.fromhex(sendData))
    recvMsg()

    cap.release()
    out.release()
    cv2.destroyAllWindows()
```

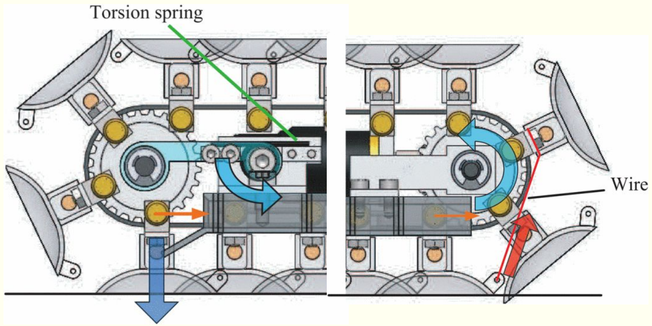
Next Steps

- Log and analyze data (this weekend hopefully)
  - Buna-N(rubber) (single-bellow) seems to be very strong, also can seal on a variety of surfaces
  - The passive/no bellow cups can stick well on a super smooth surface(acrylic) but on a normal table they barely form a seal

- Test with the vacuum/arduino/ active suction setup

- Make a decision for initial prototype (active or passive suction)
  - Begin researching, ideating, and designing the mechanism





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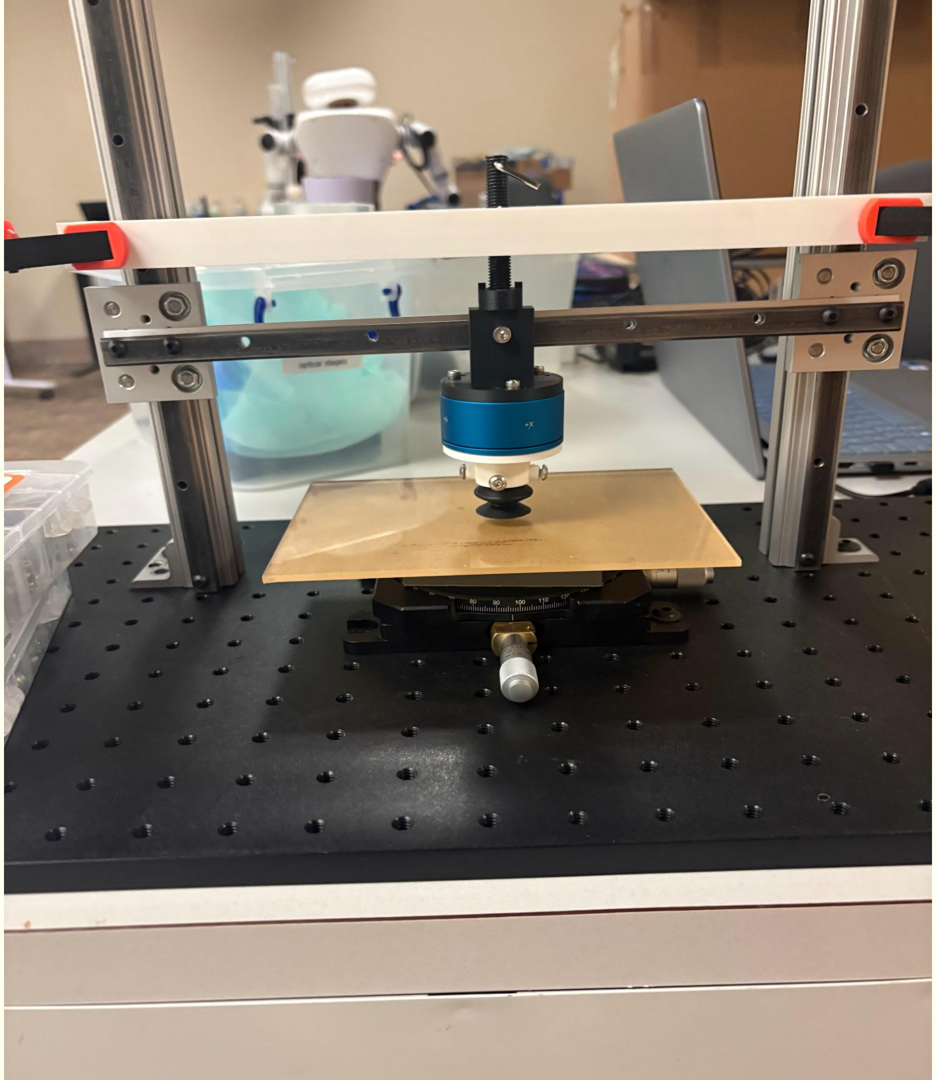
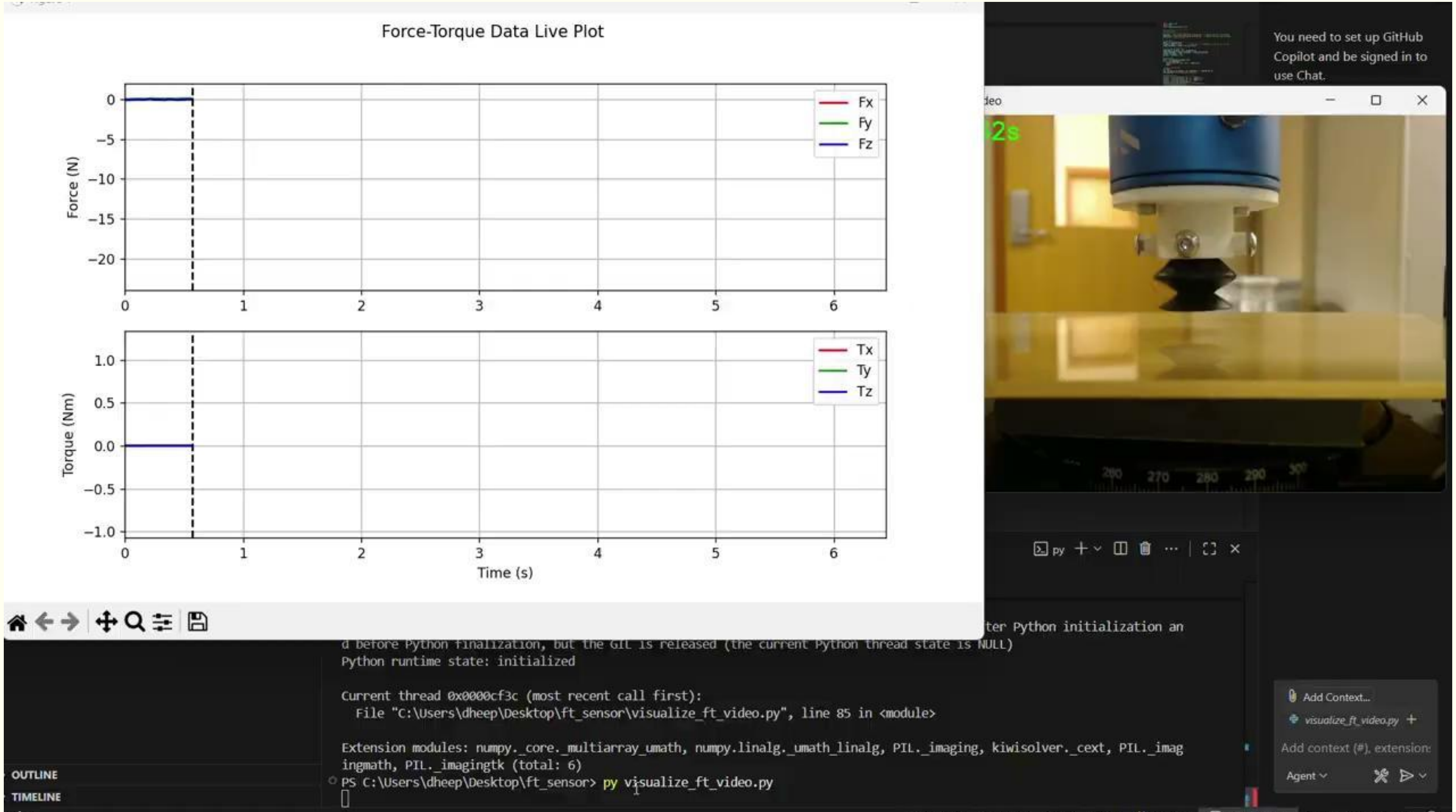
Gebelt Climbing Robot- Week 4

Testbed

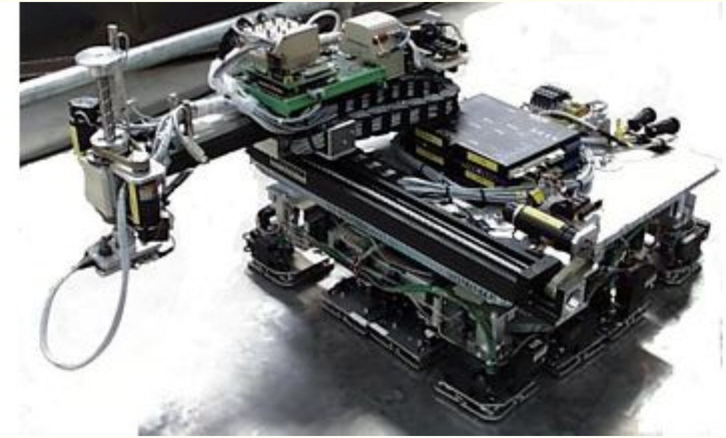


```
PS C:\Users\dheep\Desktop\ft_sensor> py max.py
File: plastic_max_ft_data_2025-10-03_14-26-37.npz -> Max Fz: 26.78 N
File: rubber_10N_ft_data_2025-10-02_23-11-08.npz -> Max Fz: 10.70 N
File: rubber_2N_ft_data_2025-10-02_22-55-26.npz -> Max Fz: 9.22 N
File: rubber_3N_ft_data_2025-10-02_23-04-32.npz -> Max Fz: 10.41 N
File: rubber_4N_ft_data_2025-10-02_23-07-05.npz -> Max Fz: 10.28 N
File: rubber_6N_ft_data_2025-10-02_23-09-20.npz -> Max Fz: 10.21 N
File: rubber_7N_ft_data_2025-10-02_23-20-44.npz -> Max Fz: 10.85 N
File: rubber_8N_ft_data_2025-10-02_23-13-28.npz -> Max Fz: 10.85 N
File: rubber_9N_ft_data_2025-10-02_23-18-30.npz -> Max Fz: 9.86 N
File: SHEAR_rubber_7N_ft_data_2025-10-02_23-28-54.npz -> Max Fz: 8.48 N
File: vinyl_plastic_2N_ft_data_2025-10-03_13-59-36.npz -> Max Fz: 21.61 N
File: vinyl_plastic_3N_ft_data_2025-10-03_14-00-36.npz -> Max Fz: 23.41 N
File: vinyl_plastic_4N_ft_data_2025-10-03_14-05-33.npz -> Max Fz: 25.86 N
File: vinyl_plastic_5N_ft_data_2025-10-03_14-03-19.npz -> Max Fz: 26.26 N
File: vinyl_plastic_6N_ft_data_2025-10-03_14-07-58.npz -> Max Fz: 25.63 N
File: vinyl_plastic_7N_ft_data_2025-10-03_14-10-56.npz -> Max Fz: 30.36 N
File: vinyl_plastic_8N_ft_data_2025-10-03_14-16-33.npz -> Max Fz: 29.33 N
File: vinyl_plastic_fail_ft_data_2025-10-03_13-41-40.npz -> Max Fz: 21.70 N
```

Logged and Visualized Data

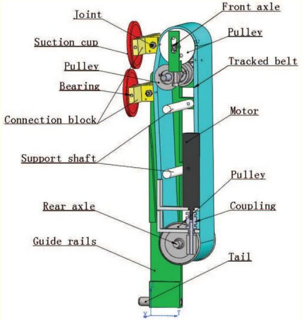


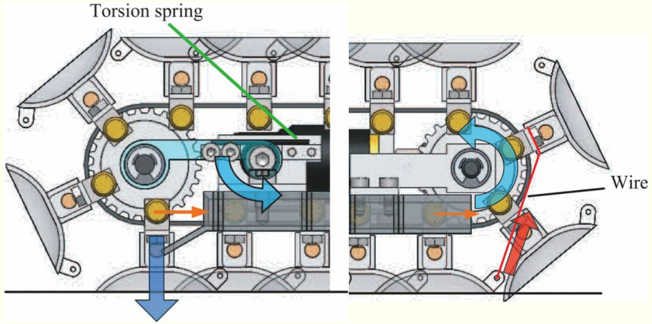
Design of a Climbing Robot for Inspecting Aircraft Wings and Fuselage



Design Considerations

- Tracked belt or torsion spring/wire
- Steering
    - Tank Drive
        - independently vary left/right track speeds (or reverse one) to turn
            - causes track skid/side-slip and higher lateral forces
            - If active suction might need to selectively adhere with specific cups
    - Center steer suction pad (a dedicated steering pad)
            - Idea: a small steerable suction pad (or pair) placed near the centerline that can produce a torque to assist turning
    - Horizontal track to move laterally and then vertical tract to move up and down





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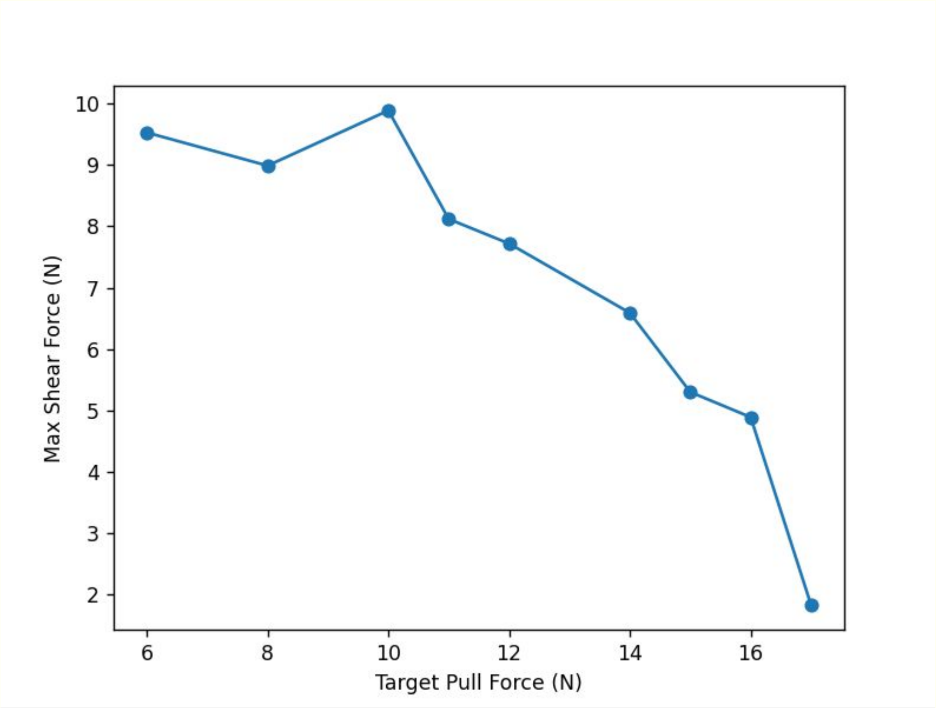
Gebelt Climbing Robot- Week 5

Shear Testing

- Measured Shear as sqrt(Fx^2 + Fy^2)
- Kept track of Fz_max as well
- Pressed with 7N every time
- Brought up to various values

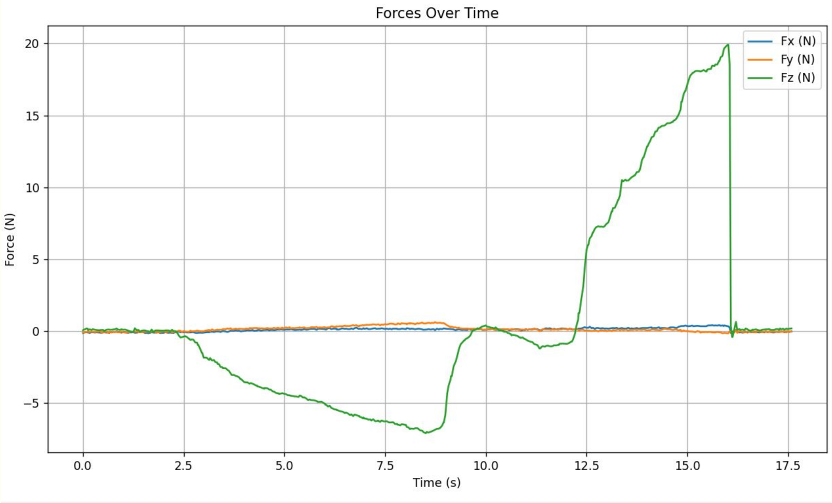
- The peak Fz values were because of continuous pulling motion
  - If we hold at say >25N it naturally pops off
  - <12N it slides so not sure if these values are valid
- Tested general max Fz and shear on various surfaces with 7N normal force
  - Back of acrylic plate (paperish material)
    - Peak Performance is less(20N), also depends on time, it naturally pops off over time
      - Ex. if i pull suction cup up 6N is will the Fz deceases to 0 -> all suction is lost within 3 seconds
  - Can’t stick to any 3D printed surface either

```
PS C:\Users\dheep\Desktop\ft_sensor> py max.py
File: SHEAR_rubber_7N_ft_data_2025-10-02_23-28-54.npz
  Max shear = 1.65 N
  Max Fz = 8.48 N

File: SHEAR_vinyl_10N_ft_data_2025-10-09_16-37-44.npz
  Max shear = 9.89 N
  Max Fz = 13.26 N

File: SHEAR_vinyl_11Nft_data_2025-10-09_14-41-01.npz
  Max shear = 8.12 N
  Max Fz = 15.18 N

File: SHEAR_vinyl_12N_ft_data_2025-10-09_14-45-14.npz
  Max shear = 7.72 N
  Max Fz = 16.27 N

File: SHEAR_vinyl_14N_ft_data_2025-10-09_15-04-31.npz
  Max shear = 6.59 N
  Max Fz = 17.73 N

File: SHEAR_vinyl_15N_ft_data_2025-10-09_16-25-45.npz
  Max shear = 5.30 N
  Max Fz = 21.39 N

File: SHEAR_vinyl_16N_ft_data_2025-10-09_14-58-21.npz
  Max shear = 4.89 N
  Max Fz = 20.60 N

File: SHEAR_vinyl_17N_ft_data_2025-10-09_16-31-11.npz
  Max shear = 1.83 N
  Max Fz = 22.14 N

File: SHEAR_vinyl_6N_ft_data_2025-10-09_16-44-16.npz
  Max shear = 9.53 N
  Max Fz = 9.19 N

File: SHEAR_vinyl_8N_ft_data_2025-10-09_16-49-22.npz
  Max shear = 8.99 N
  Max Fz = 12.89 N
```

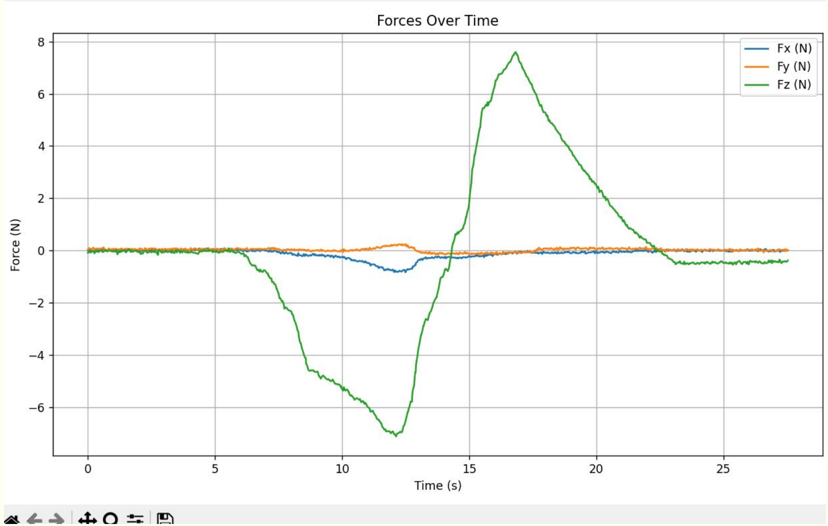


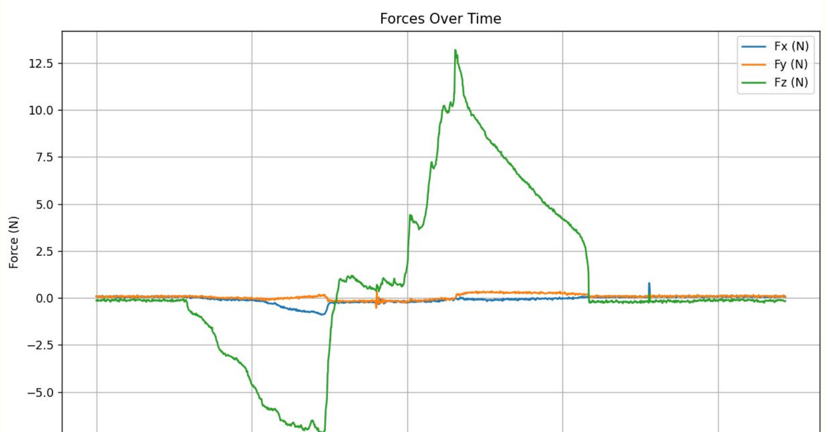
Acrylic Backing

Peak Test



Stretched and held

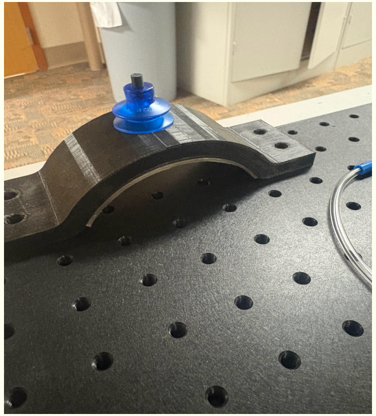


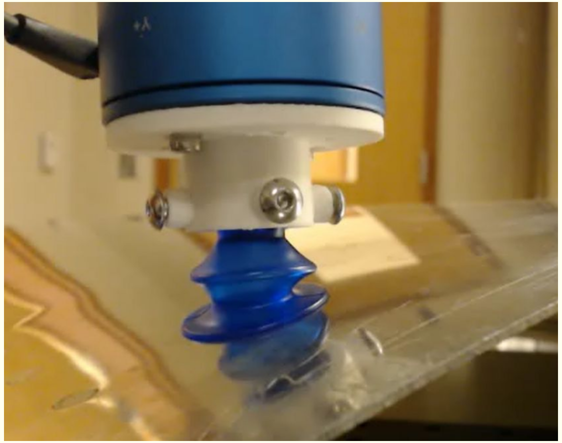


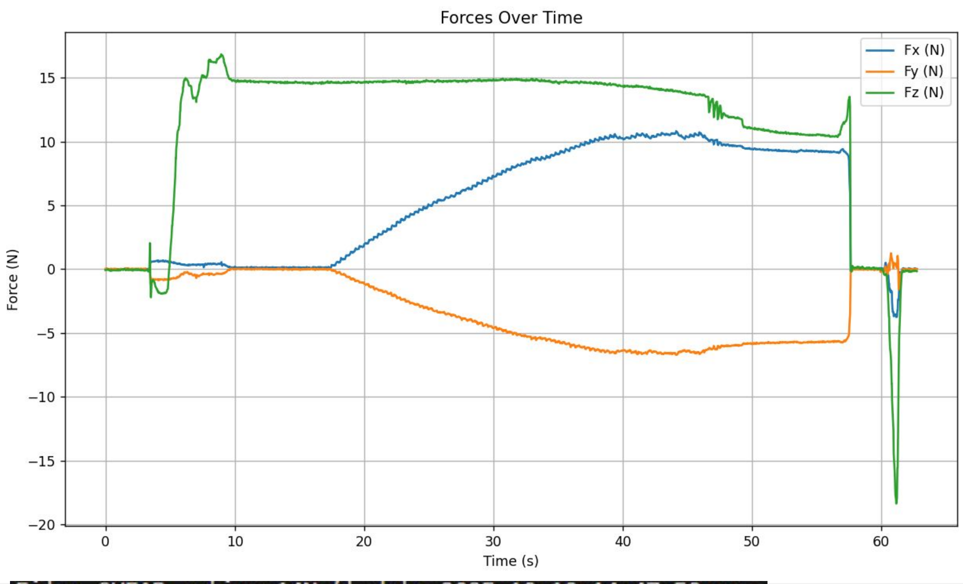
Curved Surface

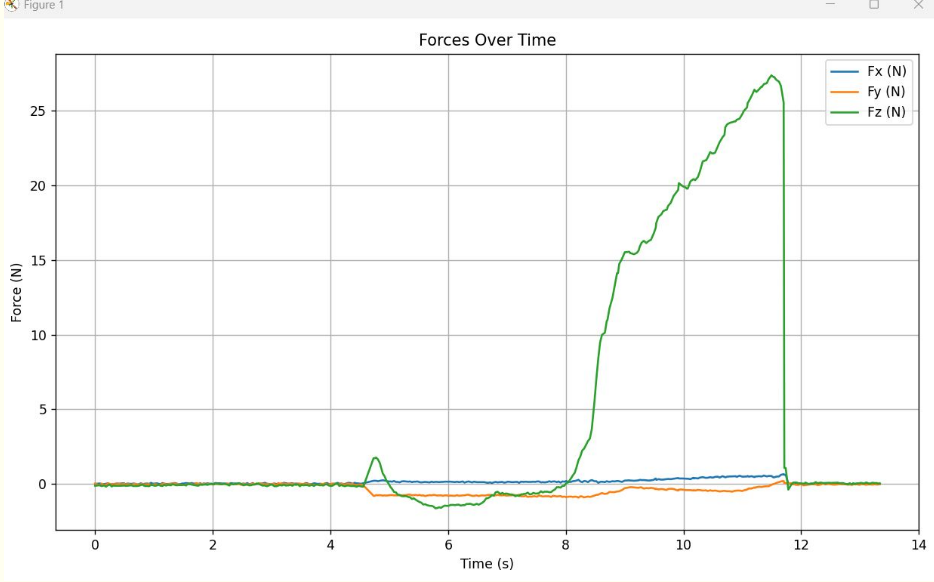
```
File: curved_vinyl_7N_ft_data_2025-10-09_23-06-45.npz
  Max Fz = 24.92 N
```

```
File: morecurved_vinyl_7N_ft_data_2025-10-09_23-17-56.npz
  Max Fz = 21.90 N
```







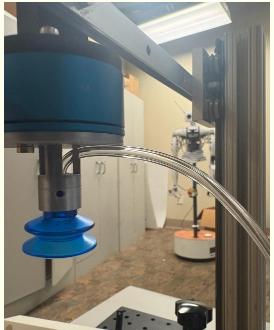



```
File: SHEAR_active_14N_ft_data_2025-10-10_14-47-50.npz
  Max shear = 12.73 N
  Max Fz = 16.83 N
```

```
File: SHEAR_vinyl_14N_ft_data_2025-10-09_15-04-31.npz
  Max shear = 6.59 N
  Max Fz = 17.73 N
```

```
File: active_test2_ft_data_2025-10-10_14-23-52.npz
  Max Fz = 27.36 N
```

- We don’t have to worry about pressing force bc it adheres by itself
- Fzmax performance is the same tho
- If we are going to blow air thru to detach it makes sense i think
- Much better performance under shear, pulled 14N, no active had max 6N shear, this had 12



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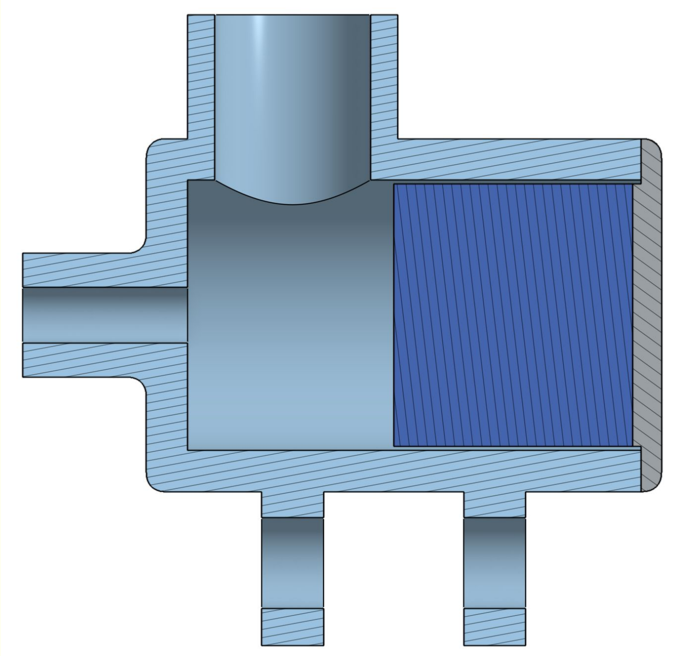
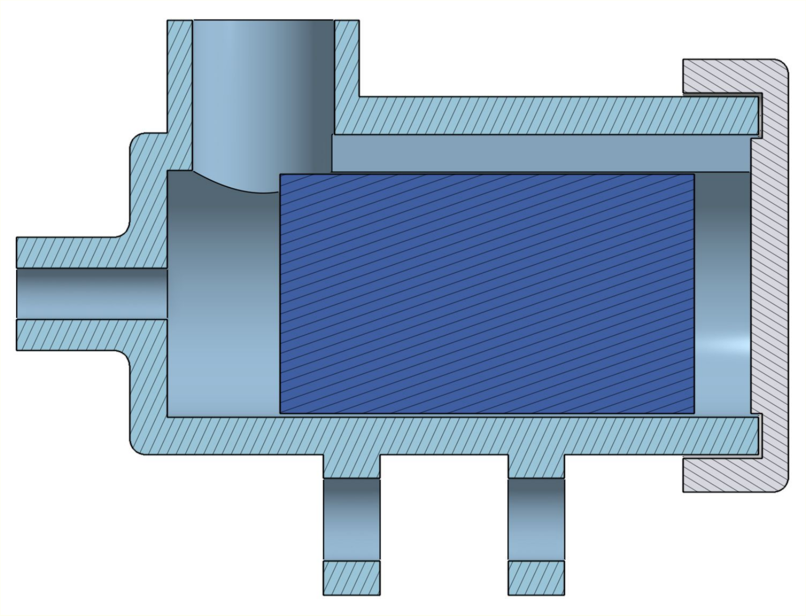
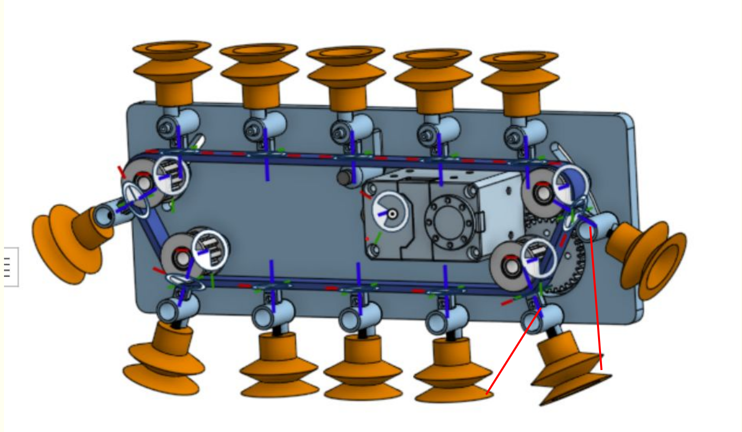
Gebelt Climbing Robot- 1/29/2026

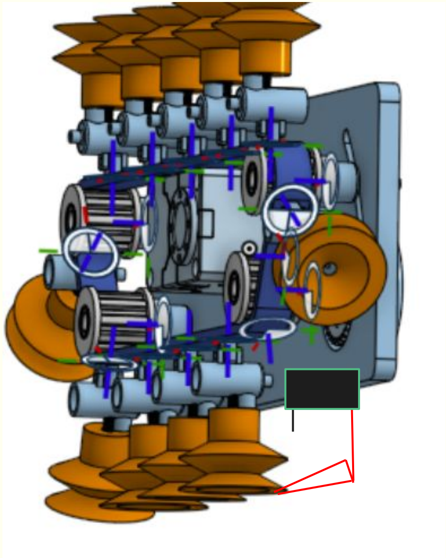




Detachment

- Pneumatic switch
- How to attach string to lip of suction cup without damaging
- Wedge on stepper motor to slide under and break seal
- Cut belt or not





Next steps

- Scope of paper
- Detachment

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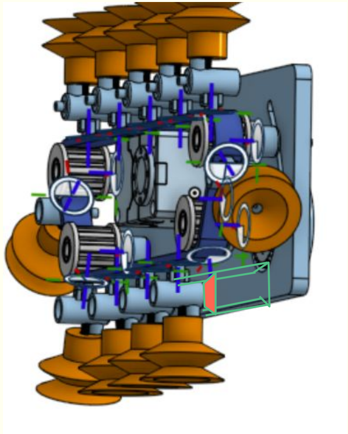
Gebelt Climbing Robot-2/11/2026

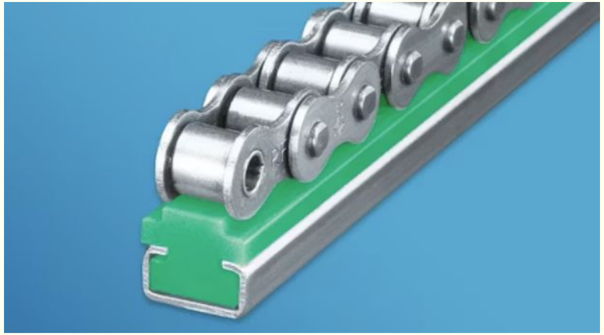
Chain Modeling

- Need chain to bear 0.46 N·m of twisting force
- Single (560lb) or double (810lb) strand
    - Physically can handle the torsion but we don’t want any displacement, chains aren't made for twisting… need to test

Chain Selection

- Could some sort of rail on inside track help for stability
  - Or valve rides a rail (helps to prevent valve connection from failing too)

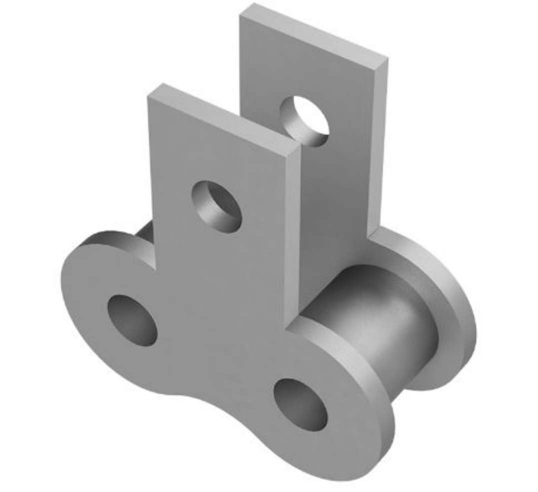




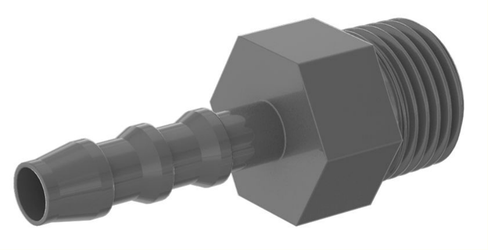
Chain Selection

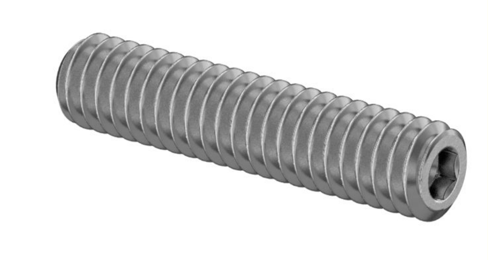
- Metal attachment is better for solid connection

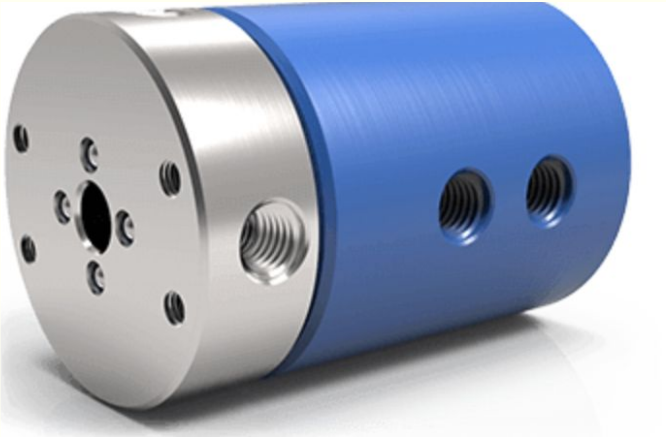
- How to support gears/pulleys sufficiently
    - Double supported shaft/sandwich
- Can probably print sprockets, and use a larger shaft/hole for more support



Pneumatics



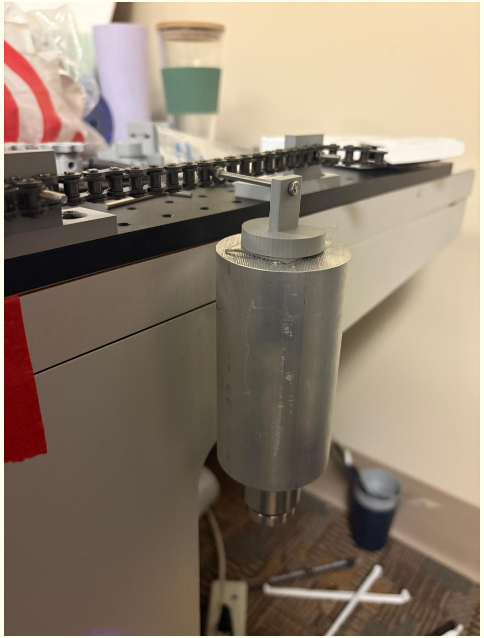




Next Steps

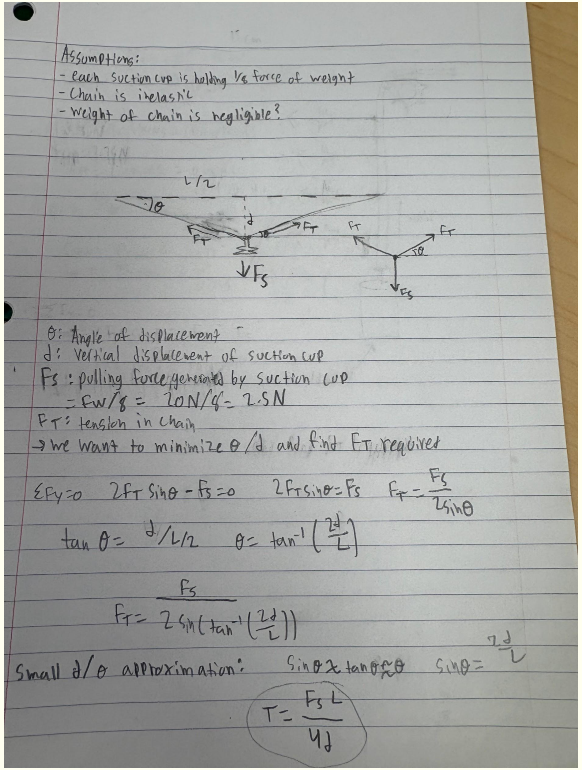
- Physical testing with the chain to ensure if it can bear the expected amount of moment and torsion
- Design documentation doc

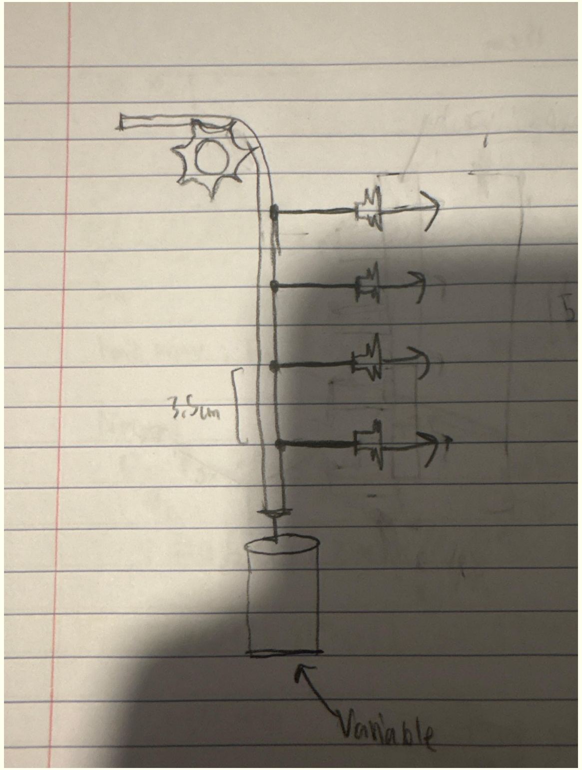
2/26/26



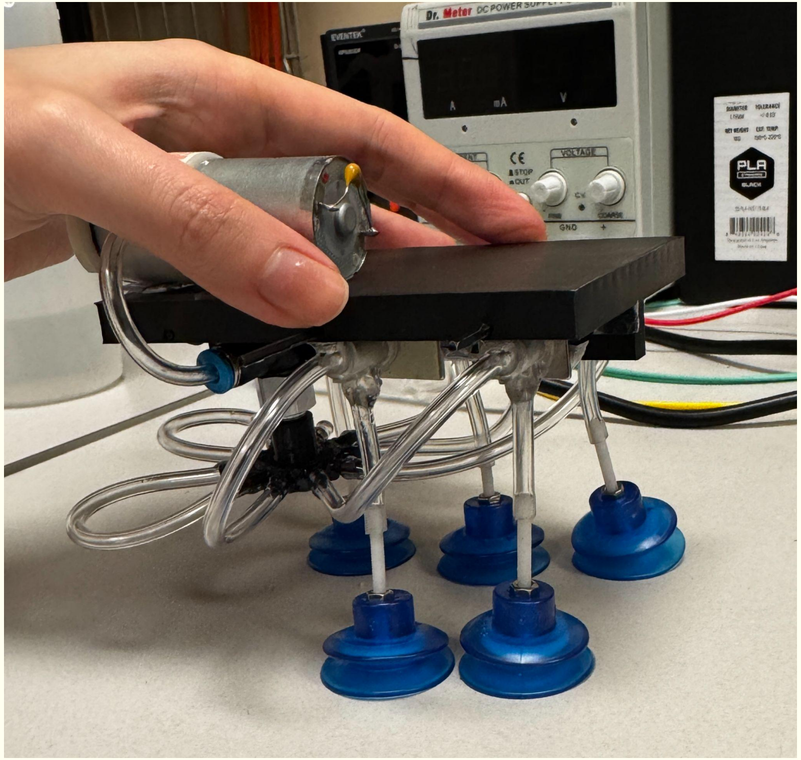
03/04/2026

Continued Testing





- 1 magnet 1 suction cup 1cm: 5.9N
- 1 magnet 5 suction cups 1 cm: 29.4N
- 1 magnet 5 suction cups 0 cm>= 49.1N



Climbingbot

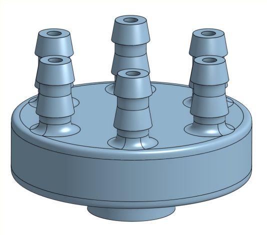
04/02/2026 Update

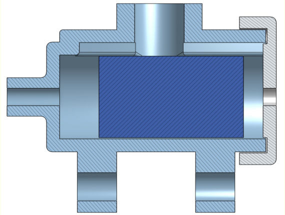
Semester Goal

- Build a system integrating electronics, suspension, gelbelt, and pneumatic subsystems and test its performance

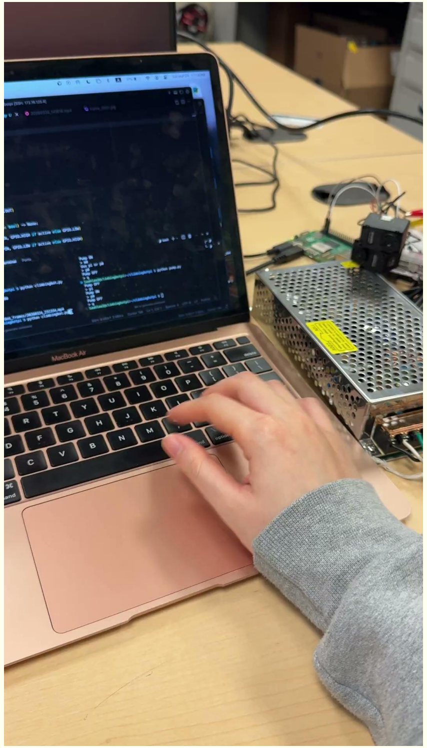
Progress – Pneumatics

System mass: 2120g
Current: 0.47A





Progress – Electronics



climbingbot.py

Dynamixel Motor Control:

m[number]   example: m100 or m0
ml[number]   left motor only
mr[number]   right motor only

Camera Recording Control:

g1 / g0   camera start / stop mp4 save

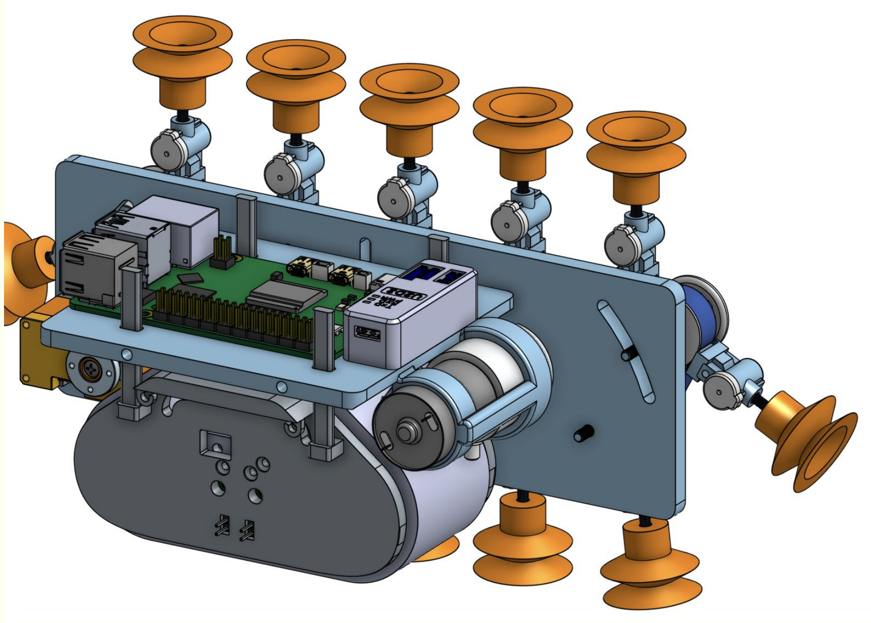
q   quit

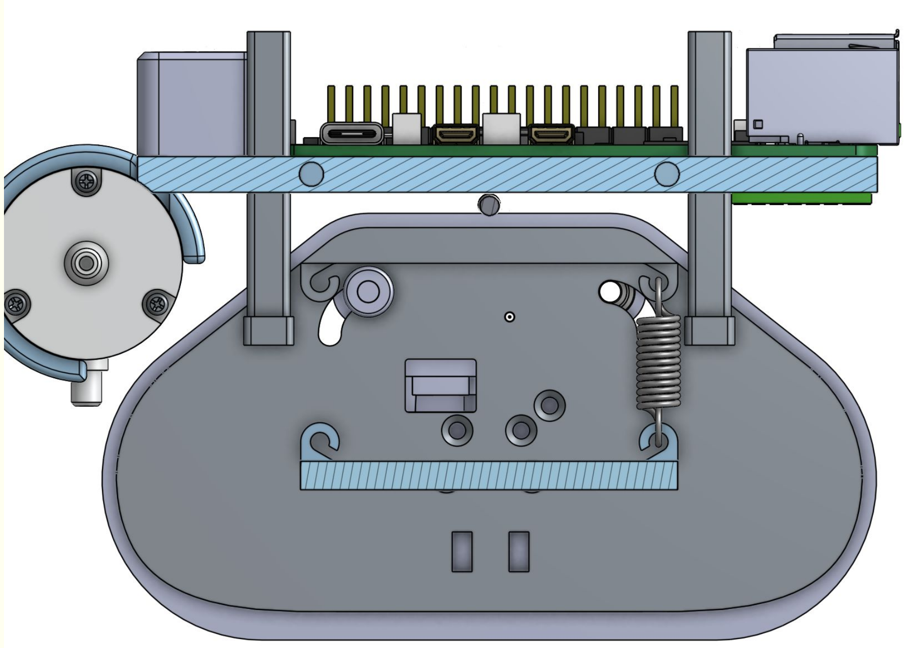
pump.py

p1 / p0   pump on / off

q   quit

Progress – Design





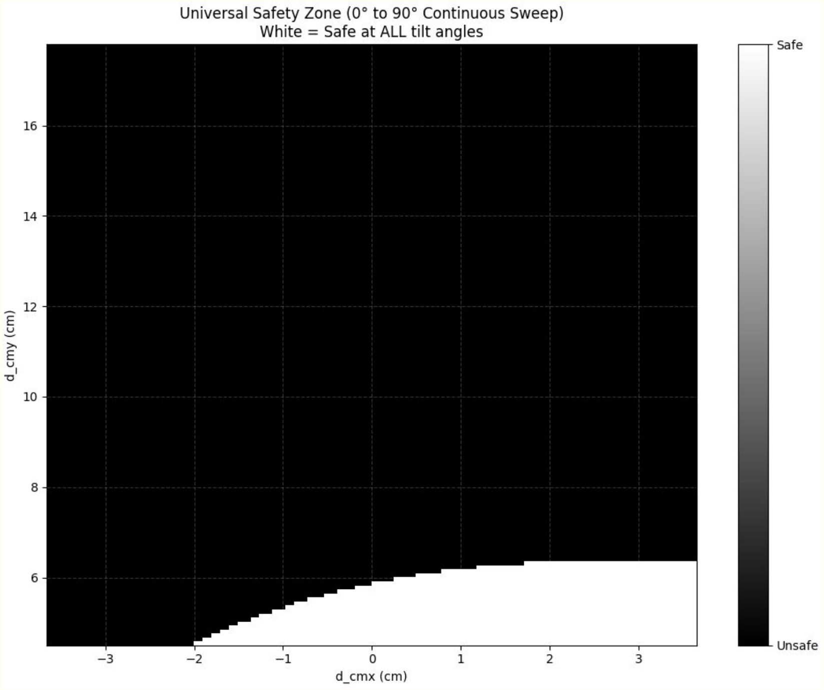
Future Plans

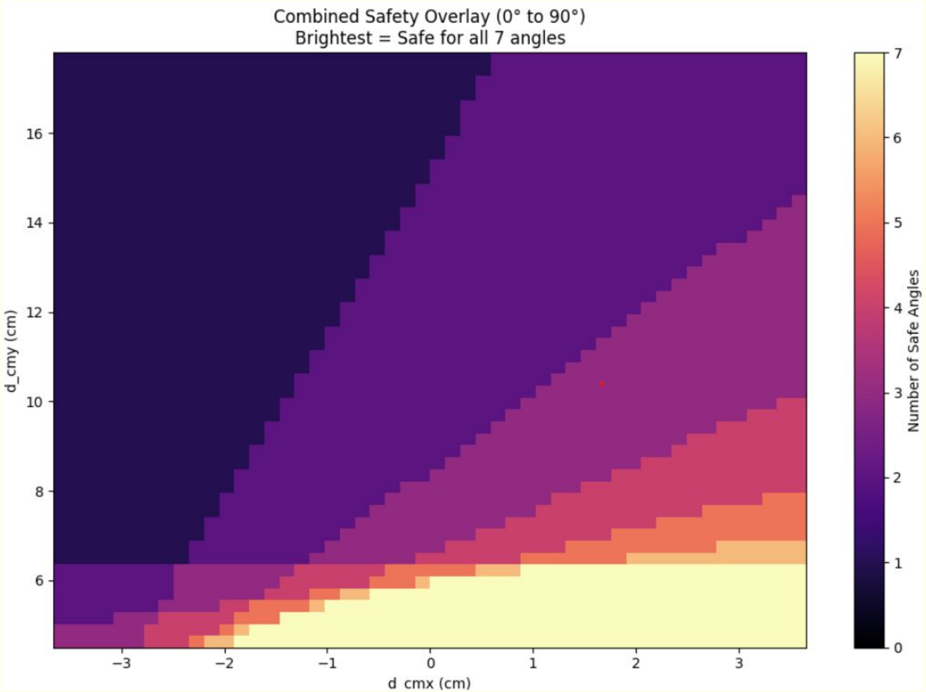
- Design finalization
- Print + Assemble + Test with electronics

Climbingbot

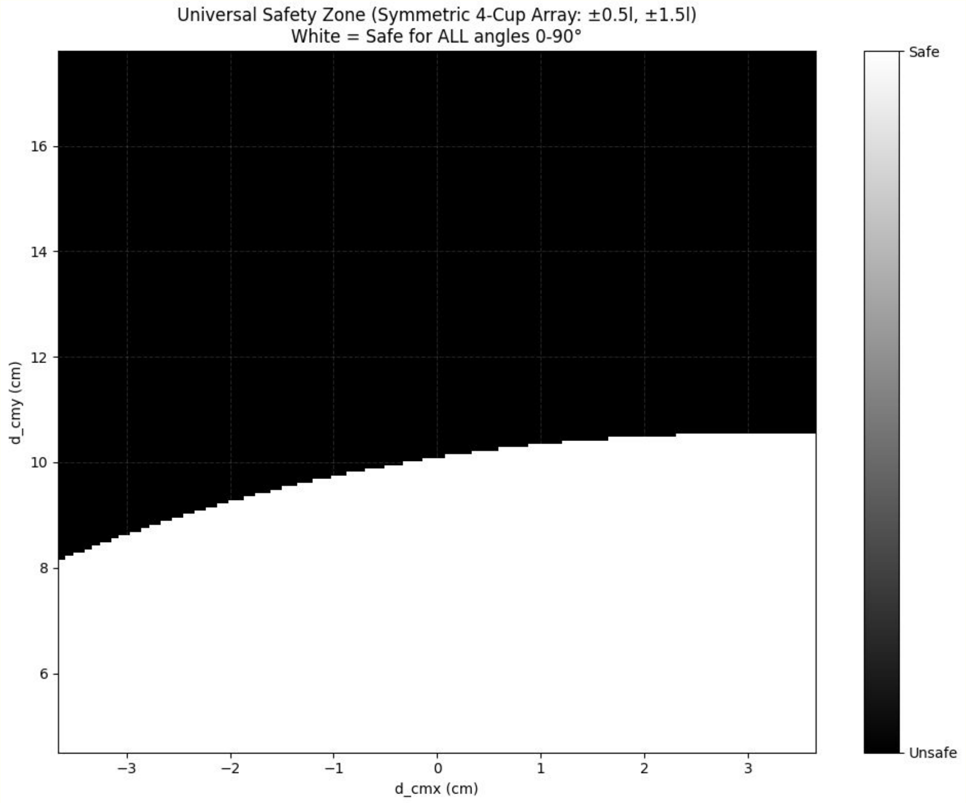
04/08/2026 Update

COM calc

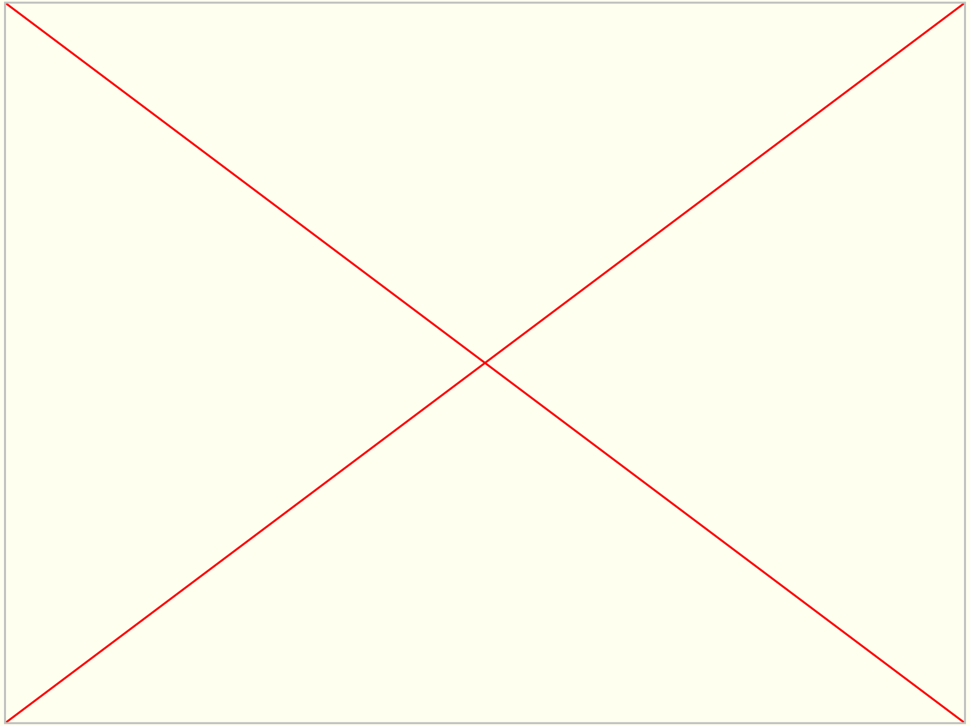

Universal Safety Zone (0° to 90° Continuous Sweep)
White = Safe at ALL tilt angles
d_cmy (cm)
d_cmx (cm)
Safe
Unsafe


Combined Safety Overlay (0° to 90°)
Brightest = Safe for all 7 angles
d_cmy (cm)
d_cmx (cm)
Number of Safe Angles

COM calc



Universal Safety Zone (Symmetric 4-Cup Array: ±0.5l, ±1.5l)
White = Safe for ALL angles 0-90°
d_cmy (cm)
16
14
12
10
8
6
−3
−2
−1
0
1
2
3
d_cmx (cm)
Safe
Unsafe

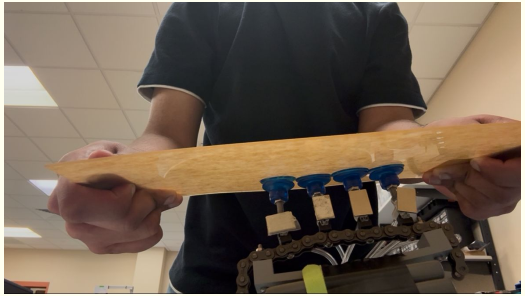


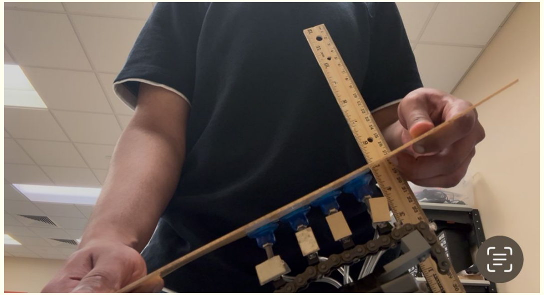
Climbingbot

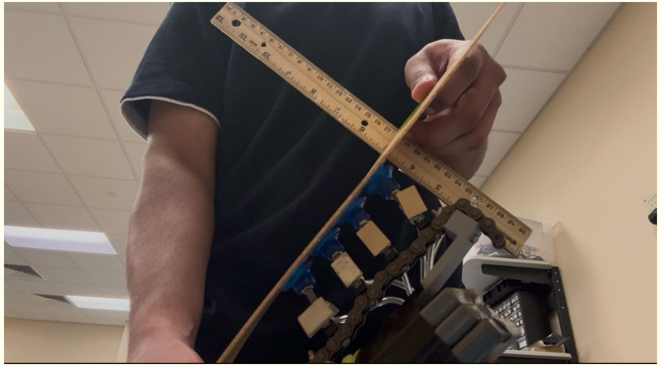
04/23/2026 Update

1min tests

- Survived 1 min tests at 0, 30, 60 degrees
- <1cm cup deformation thru these angles
  - ~0.3cm at 30 and ~0.6cm at 60
- Note: it is still very difficult to completely tension the chain and make it act as a rigid body, even as it is fixed along its length (I tried multiple times to get least tolerance), there always seems to be some slop which adds to uneven force distribution
- Weight: 15kg
- Rough COM offset ~ 3cm from chain

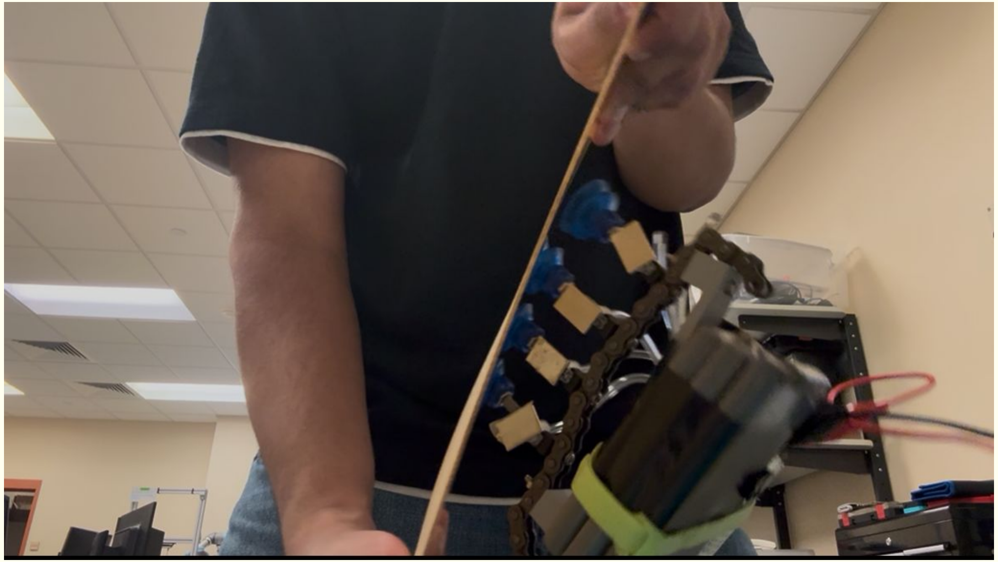






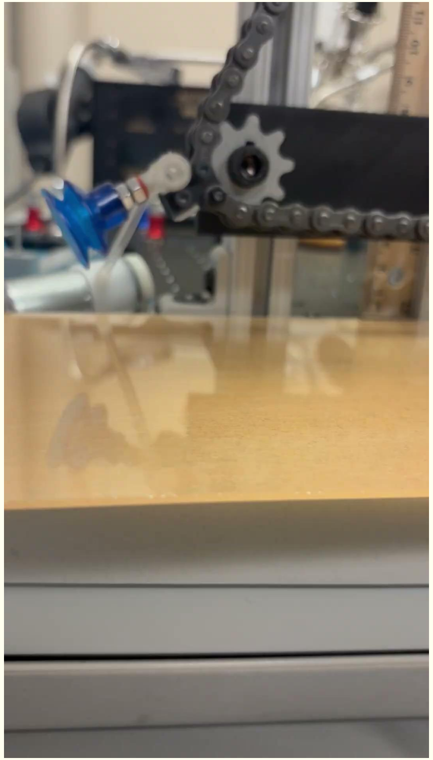
Failure

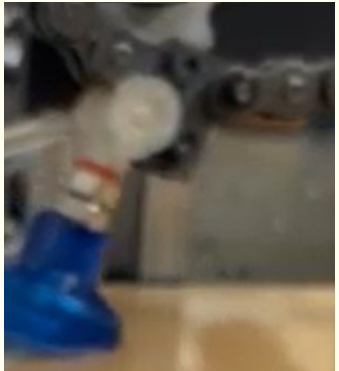
- Failed at ~75 degrees



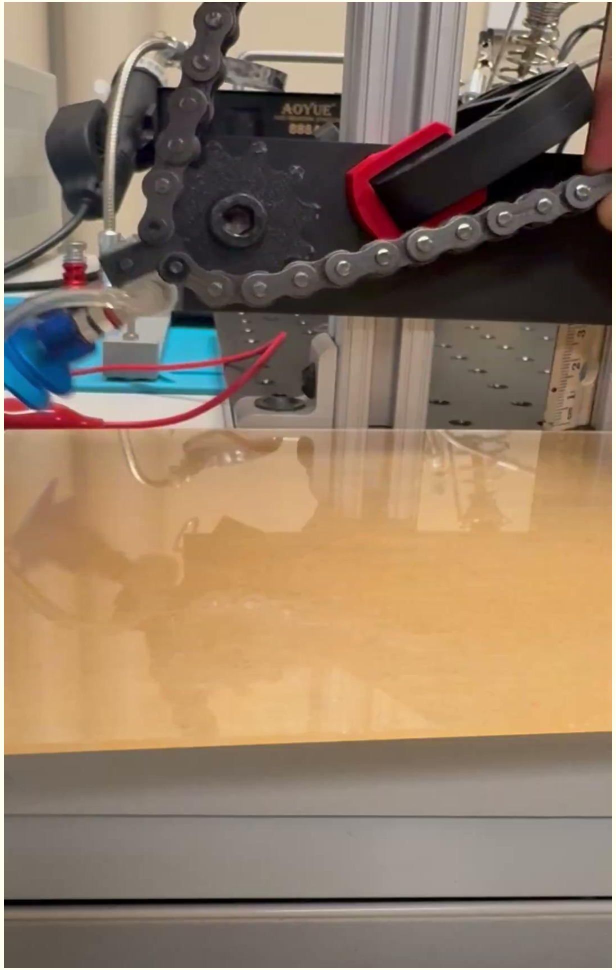
Turning radius test

- Tested using new valve design and sprockets
  - Fixed to aluminum extrusion, set at height at which the Suction cup is activated (squished height)
- Acrylic plate is left free to move, but with a 500g weight on it to add friction
- Issue #1: with new valve design since the magnet is closer to the chain it gets attracted to the chain and interferes with the connection in one direction, you can maybe see at the end of video and below
  - I am pulling very hard as it comes around, still is the chain tensioning/ rigid body issue

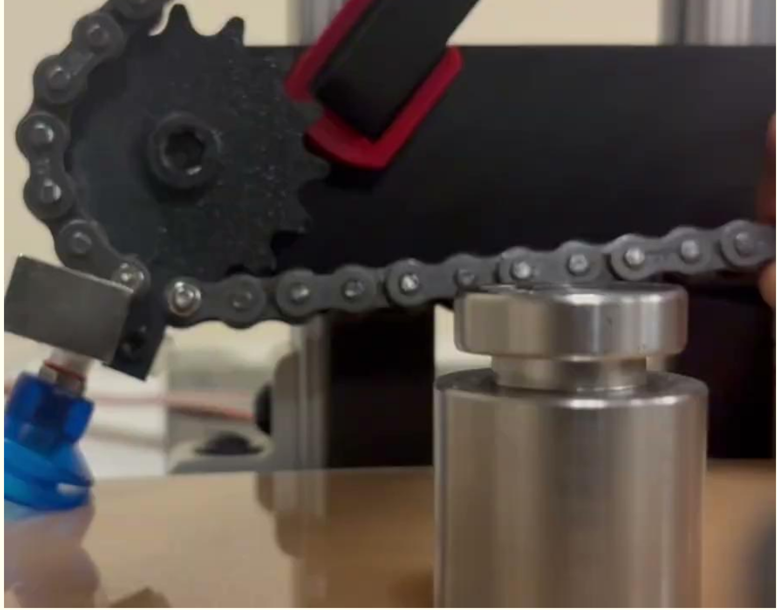


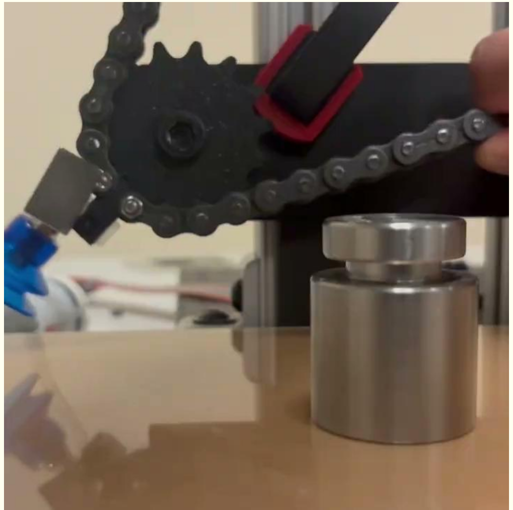


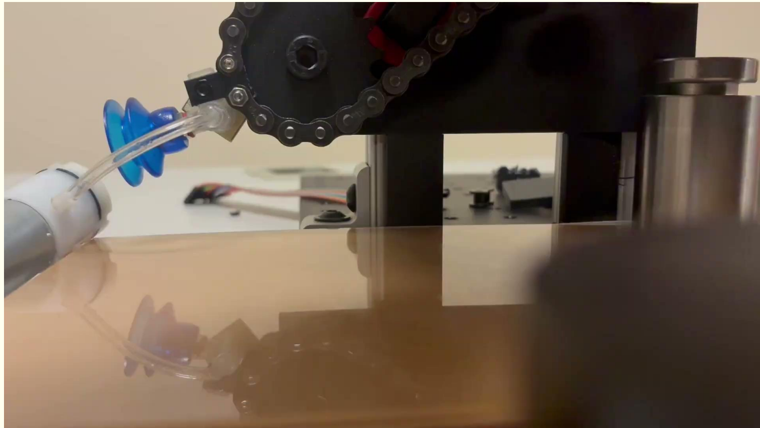
- Have tested sprocket 1.25, 1.5 inch OD, still have yet to test 1.75, 2, but maybe should figure out a better system for height and figure out valve before



- Center of sprocket is 51mm from ground
- 50.8mm OD sprocket is successful
    - With new valve design, only one direction of motion is feasible (shown in bottom right)







- Brief cad update, moved motor adjusted to new valve…